🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.8

Understand Absolute Value of Rational Numbers

Lesson 7-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

Language Objective: I can explain absolute value using the words distance, zero, opposite, and rational number.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Integer | Positive | Negative | Absolute value | Opposite | Rational number

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A whole number or its opposite, like −2, 0, or 5, is an ________.

2. A number greater than zero is ________.

3. A number less than zero is ________.

4. The distance of a number from zero, always positive, is its ________.

5. Two numbers the same distance from zero but on different sides, like 4 and −4, are ________.

6 A number you can write as a fraction of two integers, with a bottom number that is not zero, is a ______________ .

Write About the Math

How does absolute value help the rescue team compare the submarine and helicopter positions?

¿Cómo ayuda el valor absoluto al equipo de rescate a comparar las posiciones del submarino y del helicóptero?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Integer** — *Número entero*
- ☐ **Positive** — *Positivo*
- ☐ **Negative** — *Negativo*
- ☐ **Absolute value** — *Valor absoluto*
- ☐ **Opposite** — *Opuesto*

Model: *-5 and 5 are the same distance from zero because ____.* — Student says both are 5 units from zero (same absolute value) but are opposites because of their signs.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The submarine is at ____ meters, and its absolute value is ____ because ____. The helicopter is at ____ meters, and its absolute value is ____. Absolute value helps the rescue team because ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Integer
2. **Notes 2:** Positive
3. **Notes 3:** Negative
4. **Notes 4:** Absolute value
5. **Notes 5:** Opposite
6. **Notes 6:** Rational number
7. **Watch for this mistake:** *A common mistake in Integers and Absolute Value is judging size by the digits alone instead of position on the number line — students think -8 is 'more' than -3 because 8 > 3, when really -8 sits farther left (colder, deeper, or lower) and is actually the SMALLER integer. Absolute value measures distance from zero and is always positive or zero, but that distance does not decide which integer is greater — only left-to-right order on the number line does. Before you submit, ask: am I comparing distance from zero (absolute value) or position on the number line (greater/less than)?*
8. **Try It 1:** A. -5 degrees F (Ice Cave) — *On a number line, -5 is to the left of -2, so -5 is the smaller (colder) temperature.*
9. **Try It 2:** A. |-12| because 12 > 7 — *|-12| = 12 and |7| = 7. Since 12 > 7, the absolute value of -12 is greater.*